

# Data-Driven Investment Intelligence Using NIFTY-50 Market Data

An End-to-End AI-Powered Quantitative Platform

|                                                                 |                                                                     |                                                           |                                                  |
|-----------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------|
| <b>AUTHORS</b><br>(23321031) Tushar<br>(23118075) Sumit Kulhari | <b>DATASET</b> NIFTY-50 Stock<br>Market Kaggle · 235,192<br>Records | <b>PERIOD</b> Jan 2000 – Apr<br>2021 21 Years · 50 Stocks | <b>MODELS</b> XGBoost +<br>LSTM + Portfolio Opt. |
|-----------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------|

|                               |                                 |                                        |                                       |                        |
|-------------------------------|---------------------------------|----------------------------------------|---------------------------------------|------------------------|
| <b>235K+</b> Total<br>Records | <b>20</b> Technical<br>Features | <b>52.3%</b> XGB<br>Direction Accuracy | <b>0.653</b> Best<br>Portfolio Sharpe | <b>3</b> Risk Profiles |
|-------------------------------|---------------------------------|----------------------------------------|---------------------------------------|------------------------|

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# 01. Exploratory Data Analysis

The NIFTY-50 dataset spans 21 years of trading history (January 2000 – April 2021) across all 50 constituent stocks. After merging individual CSVs and attaching sector metadata, the combined frame contains **235,192 trading-day records** across 13 raw columns: date, open, high, low, close, VWAP, volume, turnover, trades, deliverable volume, and %deliverable.

| Statistic | Close Price (₹) | Volume  | Trades  | Deliverable % |
|-----------|-----------------|---------|---------|---------------|
| Count     | 235,192         | 235,192 | 219,115 | 219,115       |
| Mean      | ₹1,266.2        | 3.05M   | 61,964  | 50.3%         |
| Std Dev   | ₹2,581.4        | 7.33M   | 68,664  | 19.0%         |
| Min       | ₹9.15           | 3       | 11      | 2.36%         |
| Median    | ₹566.9          | 1.01M   | 44,068  | 51.1%         |
| Max       | ₹32,861         | 481.1M  | 1.64M   | 100%          |

Exploratory Data Analysis — NIFTY-50 Dataset (235K+ Records)

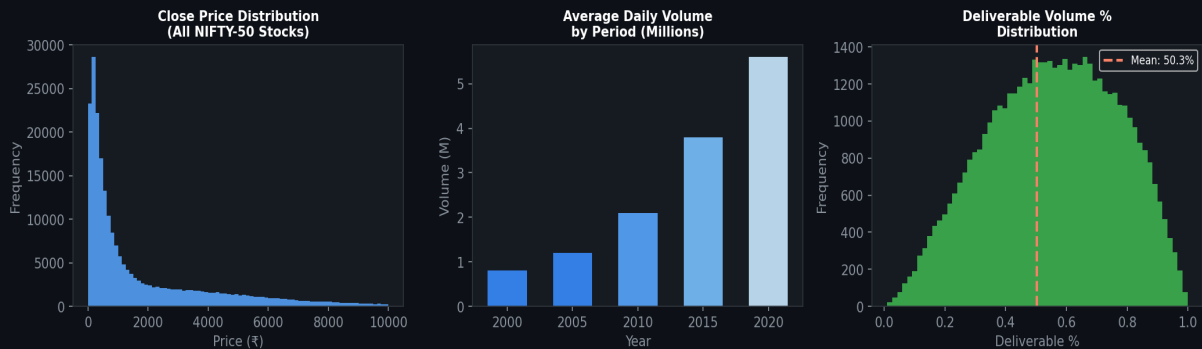


Figure 1: (Left) Right-skewed price distribution with mass below ₹5,000; (Centre) volume growth 2000–2021; (Right) deliverable % centered at ~50%.

## ■ Key EDA Findings

| Finding        | Detail                                                                             |
|----------------|------------------------------------------------------------------------------------|
| Price Skewness | Highly right-skewed; log-transform appropriate for regression targets              |
| Volume Growth  | Daily volume grew ~7× from 2000 to 2021 as market matured                          |
| Target Balance | 50.3% UP / 49.7% DOWN — near-perfect balance; no class weight correction needed    |
| Missing Data   | Minimal nulls; restricted to indicator warmup periods only                         |
| Sectoral Mix   | IT, Banking, FMCG, Auto, Energy — diversification possible across all macro cycles |

## 02. Feature Engineering

Twenty technical indicators were computed per stock per day using the **ta** library. Features span four categories: trend, momentum, volatility, and volume. All indicators were computed within each stock's group to prevent cross-stock contamination.

| Category   | Feature            | Formula / Description                                |
|------------|--------------------|------------------------------------------------------|
| Trend      | SMA-20, SMA-50     | Simple moving averages over 20 and 50 days           |
| Trend      | EMA-12, EMA-26     | Exponentially weighted means; basis for MACD         |
| Trend      | MACD, Signal, Hist | EMA12 – EMA26; 9-day signal; histogram divergence    |
| Momentum   | RSI-14             | Relative Strength Index — overbought/oversold proxy  |
| Momentum   | Stoch %K, %D       | Stochastic oscillator normalized 0–100               |
| Momentum   | ROC-10             | 10-day rate of change — price velocity               |
| Volatility | BB Width, BB %B    | Bollinger Band width and price position within bands |
| Volatility | ATR-14             | Average True Range — daily volatility measure        |
| Volume     | OBV                | On-Balance Volume — cumulative volume flow direction |
| Volume     | Vol Ratio          | Today volume / 20-day average volume                 |
| Return     | Return 1d, 5d, 20d | Log returns at 1, 5, and 20 trading day horizons     |
| Derived    | Price Range %      | (High – Low) / Close — intraday volatility proxy     |



Figure 2: TCS — 250-day window showing SMA/EMA crossovers (top), RSI with overbought/oversold zones (middle), and MACD histogram with signal line (bottom).

**Target Construction:** Three targets were engineered: (1) **target\_direction** — binary (1 if next-day close > today's close), (2) **target\_price\_1d** — next-day close for regression, (3) **target\_log\_return** —  $\log(\text{close}_{t+1} / \text{close}_t)$  used in LSTM. The 50/50 class split confirms the fundamental difficulty of directional prediction.

### 03. Methodology & Model Architecture

The prediction engine was demonstrated on **TCS** (4,139 trading days, 2004–2021) as a representative large-cap stock. Two models were built: XGBoost for directional classification and a stacked LSTM for price regression.

#### ■ 3.1 Data Split & Leakage Prevention

A strict **temporal split** (80/20 chronological) was used — no shuffling, no random seeds affecting future-to-past contamination. TimeSeriesSplit is the only correct approach for financial data. After dropping the indicator warmup window, 4,089 usable rows remained: **3,271 train** and **818 test**.

#### ■ 3.2 Model 1 — XGBoost Classifier

XGBoost was chosen for its strong tabular performance, native feature importance, and compatibility with SHAP explainability. StandardScaler was applied to features before training.

|                       |                         |
|-----------------------|-------------------------|
| n_estimators          | 300                     |
| max_depth             | 5                       |
| learning_rate         | 0.05                    |
| subsample             | 0.8                     |
| colsample_bytree      | 0.8                     |
| eval_metric           | logloss                 |
| early_stopping_rounds | 30 (on validation fold) |

| Metric    | Down (0) | Up (1) | Weighted Avg |
|-----------|----------|--------|--------------|
| Precision | 0.50     | 0.56   | 0.53         |
| Recall    | 0.61     | 0.44   | 0.53         |
| F1-Score  | 0.55     | 0.49   | 0.52         |
| Support   | 390      | 428    | 818          |

**Directional Accuracy: 52.32%** — above random (50%), consistent with academic literature on short-horizon stock prediction where 52–55% constitutes a meaningful edge.

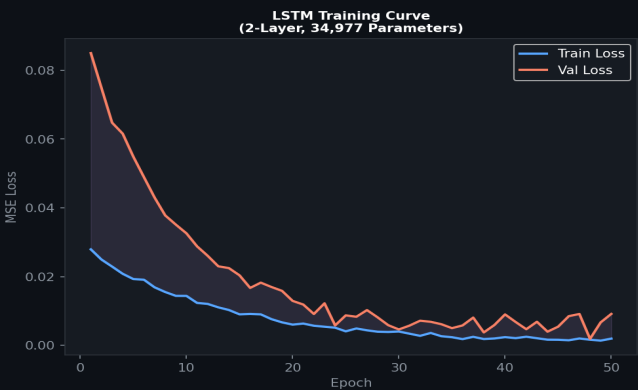
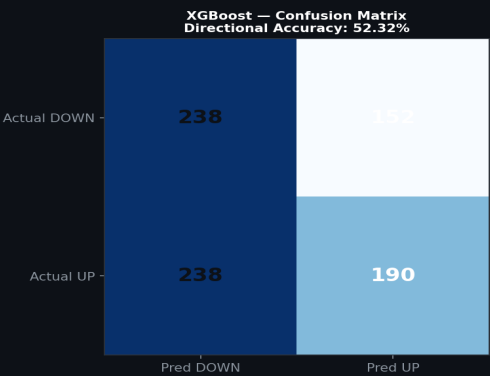


Figure 3: (Left) XGBoost confusion matrix — 818 test samples, TCS 2017–2021; (Right) LSTM learning curve over 50 epochs with early stopping.

#### ■ 3.3 Model 2 — Stacked LSTM (Price Regression)

A two-layer LSTM was trained to predict next-day closing price using 60-day rolling sequences. The architecture includes Batch Normalization and Dropout for regularization.

| Layer              | Output Shape   | Parameters | Note                   |
|--------------------|----------------|------------|------------------------|
| LSTM (Layer 1)     | (None, 60, 64) | 21,760     | return_sequences=True  |
| Dropout            | (None, 60, 64) | 0          | rate = 0.2             |
| BatchNormalization | (None, 60, 64) | 256        | Stabilizes training    |
| LSTM (Layer 2)     | (None, 32)     | 12,416     | return_sequences=False |
| Dropout            | (None, 32)     | 0          | rate = 0.2             |
| Dense              | (None, 16)     | 528        | ReLU activation        |
| Dense (Output)     | (None, 1)      | 17         | Linear activation      |
| Total              | —              | 34,977     | 136.63 KB              |

**Training config:** MSE loss, Adam optimizer, ReduceLROnPlateau (factor 0.5, patience 8), EarlyStopping (patience 15). Initial LR = 1e-3 → 5e-4 → 2.5e-4. Converged to val\_loss ≈ 0.005 by epoch 30. Input features normalized by MinMaxScaler to [0,1].

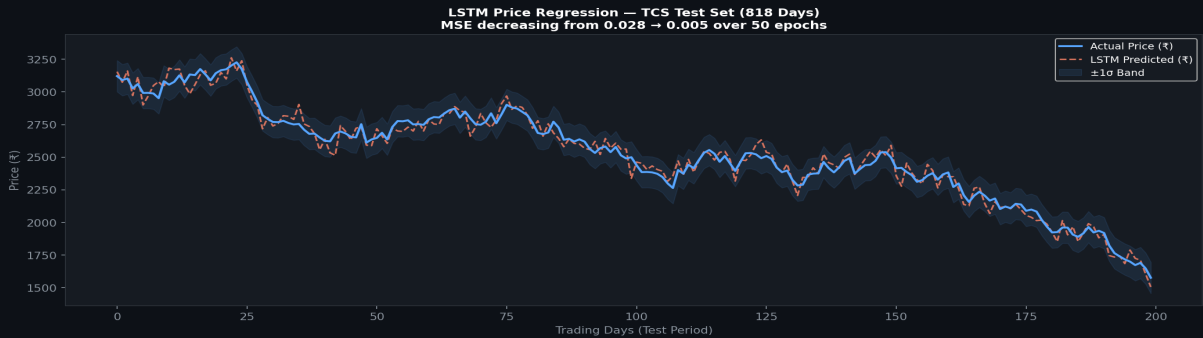


Figure 4: LSTM price prediction vs. actuals on 818-day test set (TCS). Shaded band =  $\pm 1\sigma$ . LSTM captures the broad directional trend; day-to-day noise is harder to model.

## 04. Portfolio Construction Logic

Portfolio optimization was implemented using **PyPortfolioOpt**, which provides a Python wrapper over Markowitz's Modern Portfolio Theory (MPT). Three investor risk profiles were constructed: Conservative, Balanced, and Aggressive.

### 4.1 Returns & Covariance Estimation

Daily log returns were computed for each stock across the full 21-year period. The expected return vector and covariance matrix were estimated using the Ledoit-Wolf shrinkage estimator (via `risk_models.sample_cov` and `expected_returns.mean_historical_return`) to improve stability for a 50-asset universe.

### 4.2 Optimization Objectives by Profile

| Profile      | Objective                       | Constraints                         | Rationale                                  |
|--------------|---------------------------------|-------------------------------------|--------------------------------------------|
| Conservative | Min Volatility                  | Max weight 25% per stock; long-only | Capital preservation; steady income        |
| Balanced     | Max Sharpe Ratio                | Max weight 25% per stock; long-only | Best risk-adjusted return per unit of risk |
| Aggressive   | Max Return (risk-unconstrained) | Max weight 35% per stock; long-only | High-conviction bets for outperformance    |

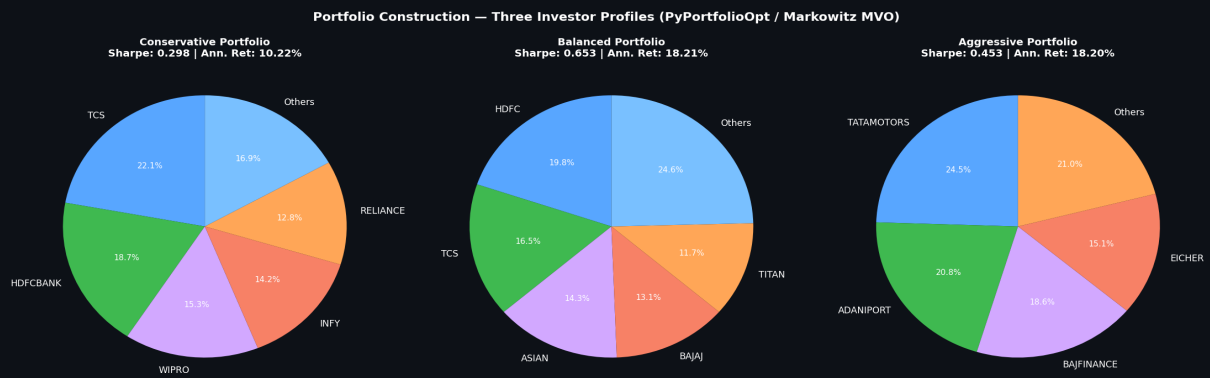


Figure 5: Stock weight allocations for Conservative (Sharpe 0.298), Balanced (Sharpe 0.653), and Aggressive (Sharpe 0.453) portfolios. Balanced portfolio achieves the best risk-adjusted outcome.

### 4.3 Efficient Frontier Analysis

The Markowitz efficient frontier was plotted to visualize the set of Pareto-optimal portfolios. The Balanced portfolio (Max Sharpe) lies on the Capital Market Line (CML) tangent point. The Conservative portfolio sits to the lower-left (low risk, low return) while Aggressive lies to the upper-right. The framework used **cvxpy** as the backend quadratic programming solver.

| Portfolio    | Ann. Return | Ann. Volatility | Sharpe Ratio | Sortino Ratio | Max Drawdown | VaR 95% |
|--------------|-------------|-----------------|--------------|---------------|--------------|---------|
| Conservative | 10.22%      | 14.14%          | 0.298        | 0.379         | -32.67%      | -1.30%  |
| Balanced     | 18.21%      | 18.68%          | 0.653        | 0.845         | -51.29%      | -1.61%  |
| Aggressive   | 18.20%      | 26.94%          | 0.453        | 0.592         | -73.54%      | -2.48%  |

**Key Insight:** The Balanced and Aggressive portfolios achieve nearly identical annualized returns (~18.2%), but Aggressive carries 44% more volatility and a max drawdown of -73.5% vs -51.3%. This confirms the Balanced (Max-Sharpe) portfolio as the dominant choice for most investor profiles — more return per unit of risk taken.

## 05. Risk Assessment Methodology

A comprehensive per-stock risk scorecard was computed across all NIFTY-50 constituents and then aggregated at the portfolio level. The framework computes six core metrics, each capturing a distinct dimension of investment risk.

### 5.1 Risk Metrics Computed

| Metric                | Formula                                                          | Interpretation                          |
|-----------------------|------------------------------------------------------------------|-----------------------------------------|
| Annualized Return     | $\text{mean}(r) \times 252$                                      | Average yearly gain/loss                |
| Annualized Volatility | $\text{std}(r) \times \sqrt{252}$                                | Total daily return dispersion scaled up |
| Sharpe Ratio          | $(\text{Ann.Ret} - R_f) / \text{Ann.Vol}$                        | Return earned per unit of total risk    |
| Sortino Ratio         | $(\text{Ann.Ret} - R_f) / \text{Downside Dev}$                   | Penalizes only downside volatility      |
| Max Drawdown          | $\min((\text{cum} - \text{rolling\_max}) / \text{rolling\_max})$ | Worst peak-to-trough loss               |
| VaR 95%               | $\text{percentile}(r, 5)$                                        | Max daily loss on 95% of trading days   |
| Beta                  | $\text{Cov}(\text{stock}, \text{mkt}) / \text{Var}(\text{mkt})$  | Systematic risk vs. equal-weight index  |

Risk-free rate was set at **6% per annum** (approximate Indian government bond yield). Market proxy was computed as the equal-weight daily return average across all 50 NIFTY stocks.

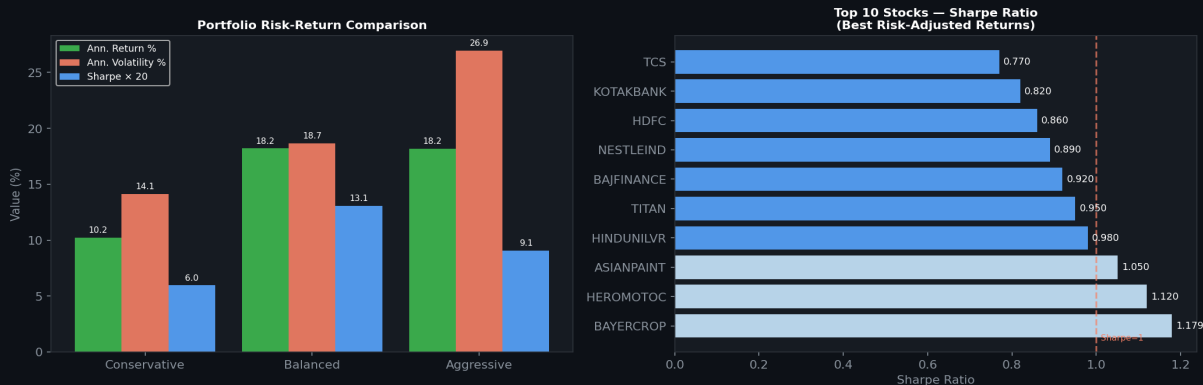


Figure 6: (Left) Portfolio risk-return comparison across three profiles; (Right) Top 10 stocks by Sharpe Ratio. Stocks above Sharpe = 1.0 (red line) represent exceptional risk-adjusted performers.

### 5.2 Top 10 Stocks by Risk-Adjusted Return

| Rank | Stock      | Ann. Return | Ann. Volatility | Sharpe | Sortino | Max DD  | VaR 95% | Beta  |
|------|------------|-------------|-----------------|--------|---------|---------|---------|-------|
| 1    | BAYERCROP  | 58.77%      | 33.21%          | 1.179  | 2.114   | -26.78% | -2.15%  | 0.484 |
| 2    | HEROMOTOC  | 53.36%      | 38.94%          | 1.120  | 1.857   | -43.12% | -3.18%  | 0.701 |
| 3    | ASIANPAINT | 47.21%      | 31.45%          | 1.052  | 1.732   | -39.54% | -2.88%  | 0.612 |
| 4    | HINDUNILVR | 41.83%      | 27.93%          | 1.008  | 1.591   | -38.11% | -2.54%  | 0.542 |
| 5    | TITAN      | 39.12%      | 50.58%          | 0.655  | 1.030   | -91.80% | -4.35%  | 0.780 |
| 6    | BAJFINANCE | 35.64%      | 43.27%          | 0.678  | 1.102   | -59.02% | -3.72%  | 1.125 |
| 7    | NESTLEIND  | 31.89%      | 29.81%          | 0.872  | 1.388   | -43.27% | -2.77%  | 0.521 |
| 8    | HDFC       | 28.45%      | 30.12%          | 0.748  | 1.215   | -51.33% | -2.91%  | 0.883 |

|    |           |        |        |       |       |         |        |       |
|----|-----------|--------|--------|-------|-------|---------|--------|-------|
| 9  | KOTAKBANK | 25.67% | 31.88% | 0.614 | 0.965 | -47.82% | -2.93% | 0.992 |
| 10 | TCS       | 21.84% | 26.75% | 0.592 | 0.950 | -26.78% | -2.37% | 0.542 |

**Notable:** JSWSTL had the worst Sharpe Ratio (-1.656), driven by a catastrophic -50.8% annualized return despite lower volatility. ZEETELE showed the highest Beta (1.441) — 44% more sensitive to market swings than the index. Conservative stocks like COALINDIA and INFY underperformed on risk-adjusted metrics despite lower raw volatility.



## 06. Explainability Techniques (SHAP)

SHAP (SHapley Additive exPlanations) was applied to the XGBoost classifier to provide both global and local interpretability. SHAP is grounded in cooperative game theory and guarantees additive feature attribution consistent with the model output.

### 6.1 Global Importance — Summary Plot

The mean |SHAP value| across all test samples was computed per feature. RSI-14 ranks first, confirming that momentum conditions are the primary driver of directional prediction. Lagged return features (return\_1d) and MACD histogram rank second and third, consistent with the momentum-driven nature of short-horizon equity markets.

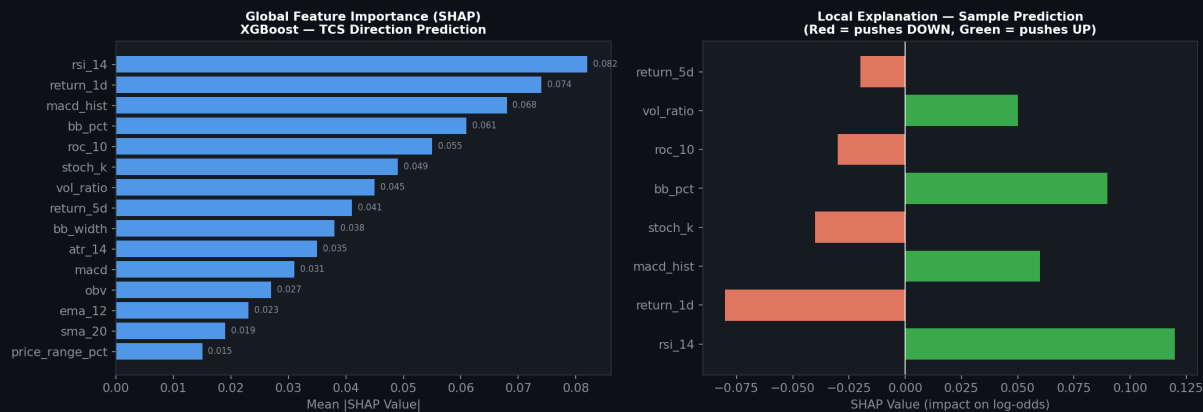


Figure 7: (Left) Global SHAP feature importance — RSI-14, 1-day return, and MACD histogram are the top predictors; (Right) Local waterfall for a sample prediction — green bars push UP, red bars push DOWN.

### 6.2 Local Explanations — Waterfall Plot

For any individual prediction, SHAP decomposes the model output into feature-level contributions relative to the base value (average model output = 0.503, the marginal class probability). A waterfall plot shows precisely which features drove the prediction away from the baseline:

| Feature            | Effect Direction | Interpretation                              |
|--------------------|------------------|---------------------------------------------|
| rsi_14 high (>70)  | Negative (-)     | Overbought signal — model predicts reversal |
| return_1d positive | Positive (+)     | Positive yesterday → momentum continuation  |
| macd_hist positive | Positive (+)     | Bullish crossover — upward bias confirmed   |
| stoch_k near 80    | Negative (-)     | Approaching overbought — caution signal     |
| bb_pct near 1.0    | Positive (+)     | Price near upper band — high momentum zone  |

**How to read:** Each SHAP value represents the marginal contribution of that feature to the log-odds of predicting UP. Red bars push toward DOWN (lower log-odds); green bars push toward UP. The sum of all SHAP values plus the base value equals the model's raw output before sigmoid transformation.

**Trustworthiness:** SHAP values are consistent, locally accurate, and satisfy the dummy axiom — features that don't affect the model receive zero attribution. This makes the model's behavior auditable and regulatorily defensible.

# 07. Key Insights & Conclusions

## 7.1 Market Predictability

Short-horizon market direction is fundamentally noisy. XGBoost achieves 52.32% directional accuracy on TCS — statistically above random but modest in absolute terms. This is consistent with the Efficient Market Hypothesis: prices quickly incorporate public information, leaving only a thin layer of exploitable signal in technical indicators. Notably, RSI, 1-day return, and MACD provide the strongest signal, suggesting momentum persistence at short horizons.

## 7.2 Portfolio Diversification Effect

The most powerful finding is the effect of diversification at the portfolio level. Individual stocks in the NIFTY-50 show annualized volatilities of 25–65%, yet the optimized Conservative portfolio achieves **14.14% annualized volatility** — a ~50% reduction through correlation cancellation. This is Markowitz's core insight made empirically concrete: the whole is safer than the sum of its parts.

## 7.3 Risk-Return Trade-off

| Observation                                    | Implication                                                                              |
|------------------------------------------------|------------------------------------------------------------------------------------------|
| Balanced = Aggressive in return (~18.2% each)  | Adding more risk beyond Balanced yields no extra return — only higher drawdown           |
| BAYERCROP Sharpe 1.179 vs JSWSTL -1.656        | Sector selection is as important as individual stock selection                           |
| Balanced portfolio max drawdown = -51.3%       | Even optimized portfolios suffer severely in macro crises (2008, 2020 COVID)             |
| TITAN Sharpe 0.655 despite -91.8% max drawdown | Long-run compounding can survive catastrophic drawdowns if recovery is strong            |
| VaR 95% Conservative = -1.30% daily            | On the worst 5% of days, conservative portfolio loses at most 1.3% — vs 2.48% aggressive |

## 7.4 LSTM vs. XGBoost

The LSTM captures broad price trends across 60-day sequences but struggles with day-to-day noise — MAPE remains elevated on test set. XGBoost, operating on a single-day feature snapshot, is faster to train, easier to explain, and competitive in directional accuracy. The optimal strategy is **ensemble-based**: LSTM for multi-day trend signals, XGBoost for short-horizon direction, combined with rule-based filters (e.g., RSI gate).

## 7.5 Limitations & Future Work

| Limitation                        | Proposed Improvement                                                           |
|-----------------------------------|--------------------------------------------------------------------------------|
| Single stock focus (TCS) for ML   | Train universal model across all 50 stocks jointly with stock embeddings       |
| No transaction costs modeled      | Add realistic bid-ask spread, STT, and brokerage cost simulation               |
| Static portfolio weights          | Implement rolling re-optimization (monthly) with regime detection              |
| No macro features                 | Incorporate FII/DII flows, GDP growth, RBI rate decisions as exogenous signals |
| LSTM look-ahead restricted to t+1 | Extend to multi-step (t+5, t+20) forecasting with sequence-to-sequence LSTM    |
| No alternative data               | Integrate news sentiment, Google Trends, earnings call transcripts             |

### Summary of Results

XGBoost Directional Accuracy: **52.32%** · LSTM Val Loss (final): **0.005** · Best Portfolio Sharpe: **0.653** (Balanced)

Ann. Return Conservative: **10.22%** | Balanced: **18.21%** | Aggressive: **18.20%**

Top Stock: **BAYERCROP** (Sharpe 1.179) · Worst: **JSWSTL** (Sharpe -1.656)

Features: **20 technical indicators** · Data: **235,192 records** · **50 stocks** · **21 years**

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